

Final Report

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Abstract

This report investigates whether ARIMA models can accurately forecast Singapore's total fertility rate (TFR) and total live births (TLB) while capturing short-term fluctuations. Annual data from 1960 to 2024 were obtained from the Singapore Department of Statistics and split into a training period (1960-2012) and a testing period (2013-2024). A range of ARIMA models were developed using combinations of logarithmic transformation and differencing to achieve stationarity, with model selection based on Akaike's Information Criterion (AIC), corrected AIC (AICc), and forecast accuracy measures including mean squared error (MSE) and mean absolute error (MAE). The results show that ARIMA models effectively capture long-term declining trends in both TFR and TLB but less effective in capturing short-term fluctuations. Forecast performance varied depending on model specification, with no single model performing best across all evaluation criteria. Overall, while ARIMA models are suitable for long-term forecasting, their reliance on historical data limits their ability to capture structural and external influences, reducing their reliability under changing demographic conditions.

1. Introduction

Total fertility rates (TFRs) are key population indicators that are widely monitored because they influence age structure and population growth (Wilmoth et al., 2022; World Health Organization). Globally, fertility rates have declined over recent decades (Chen et al., 2018; Shair et al., 2024; Vanella, 2024). In Singapore, the total fertility rate has dropped below the replacement level required for a population to replace itself across generations (Jones & Hamid, 2015; Tan, 2024; Wilmoth et al., 2022). Therefore, forecasting TFR and total live births (TLB) may provide insight into future fertility patterns and population change in Singapore. Preliminary exploratory data analysis indicated a long-term decline in TFR and TLB with periodic fluctuations, suggesting the use of ARIMA models.

Data from 1960 to 2024 on TLB and TFR were obtained from the Singapore Department of Statistics. (2023). The research question addressed in this report is: Can ARIMA models accurately forecast Singapore's TLB and TFR while capturing short-term fluctuations?

This report is structured as follows. Section 2 discusses the fertility decline in Singapore. Section 3 describes the process of developing and selecting appropriate ARIMA models. Section 4 presents the forecasting results and model performance based on AIC, AICc, MSE and MAE for the chosen viable models. Section 5 discusses the findings and limitations. Section 6 concludes the report.

2. Literature Review

Around 1965, Singapore introduced policies aimed at reducing population growth, which contributed to subsequent declines in fertility rates (Chen et al., 2018; Goodkind, 1996; Singh et al., 1988; Vanella, 2024). Despite this long-term decline, there may be periodic fluctuations in TFR and TLB. These fluctuations may be associated with cultural preferences surrounding Lunar zodiac Dragon years (1964, 1976, 1988, 2000, 2012 and 2024), which are considered especially auspicious and have been associated with temporary increases in fertility rates and live births (Goodkind, 1996; Jones & Hamid, 2015). Forecasting such series therefore requires models that can capture both long-term trends and short-term variation. Among the various models for time series forecasting, ARIMA models are particularly flexible because they allow the model structure to be selected according to the characteristics of the data rather than imposing a single model form (Kolambe & Arora, 2024; Newbold, 1983). Consequently, ARIMA models were selected as the forecasting approach for Singapore's TFR and TLB.

3. Methods

For both TFR and TLB, an identical modelling workflow was applied. The analysis focused on the period from 1960-2012 for model fitting, with years 2013-2024 reserved for evaluating forecast accuracy. Initial plots were used to examine the presence of trends, seasonality, and other notable patterns, which indicated a long-term decreasing trend in both series. Several transformations, including logarithmic transformation and first- and second-order differencing were applied to achieve stationarity. Stationarity was assessed using the KPSS test. For each transformed series, autocorrelation (ACF) and partial autocorrelation (PACF) plots were used to identify candidate ARIMA orders. Candidate ARIMA models were fitted, and residual ACF and PACF plots were examined to determine whether the residuals resembled white noise. Forecast accuracy was evaluated using mean squared error (MSE) and mean absolute error (MAE) while model selection was guided by Akaike's Information Criterion (AIC) and the corrected Akaike's Information Criterion (AICc). This procedure was applied consistently to both series to ensure comparability of results. All ACF and PACF plots, along with model outputs, are provided in the Appendix.

4. Results

This section presents model fitting and forecast accuracy for TFR and TLB.

Table 1. TFR - no log transformation

Model	AIC	AICc	MSE	MAE
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547

Table 2. TFR - log transformed

Model	AIC	AICc	MSE	MAE
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933

For TFR, models fitted to both non-log and log-transformed series were compared using AIC, AICc, MSE, and MAE. The ARIMA(13,2,2) model achieved the lowest forecast error and was selected as the preferred model.

Table 3. TLB - no log transformation

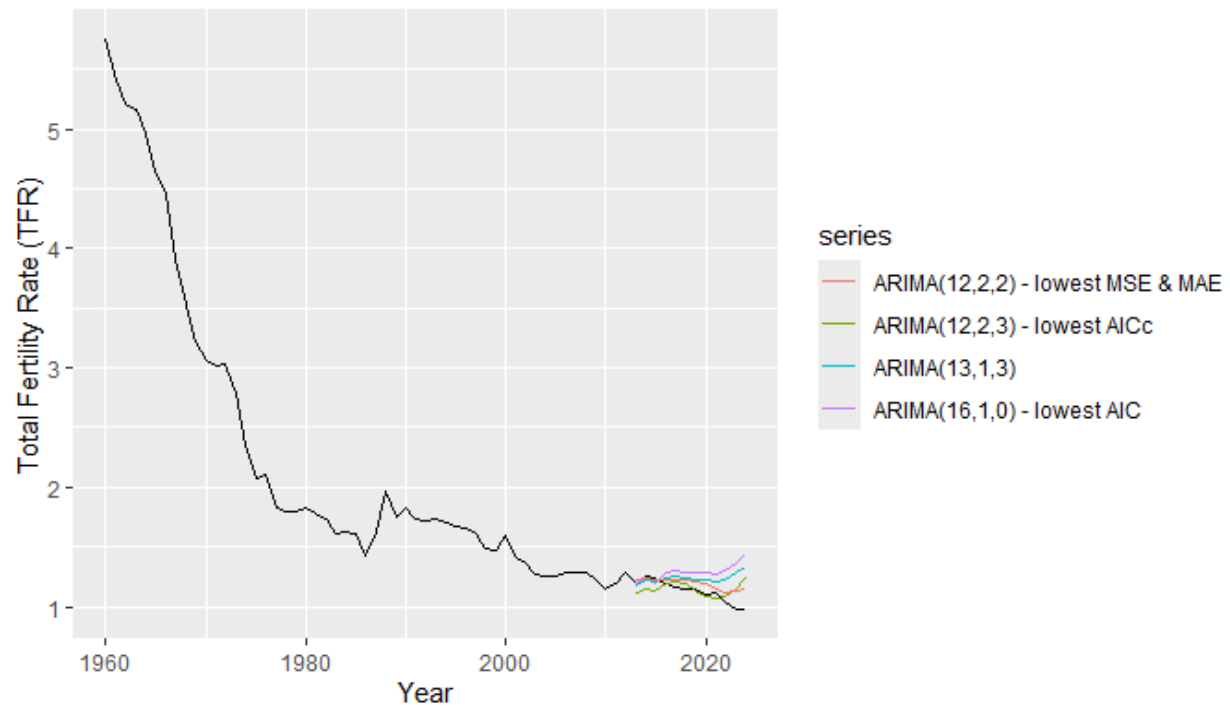
Model	AIC	AICc	MSE	MAE
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802

Table 4. TLB - log transformed

Log Transformed TLB Model	AIC	AICc	MSE	MAE
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92

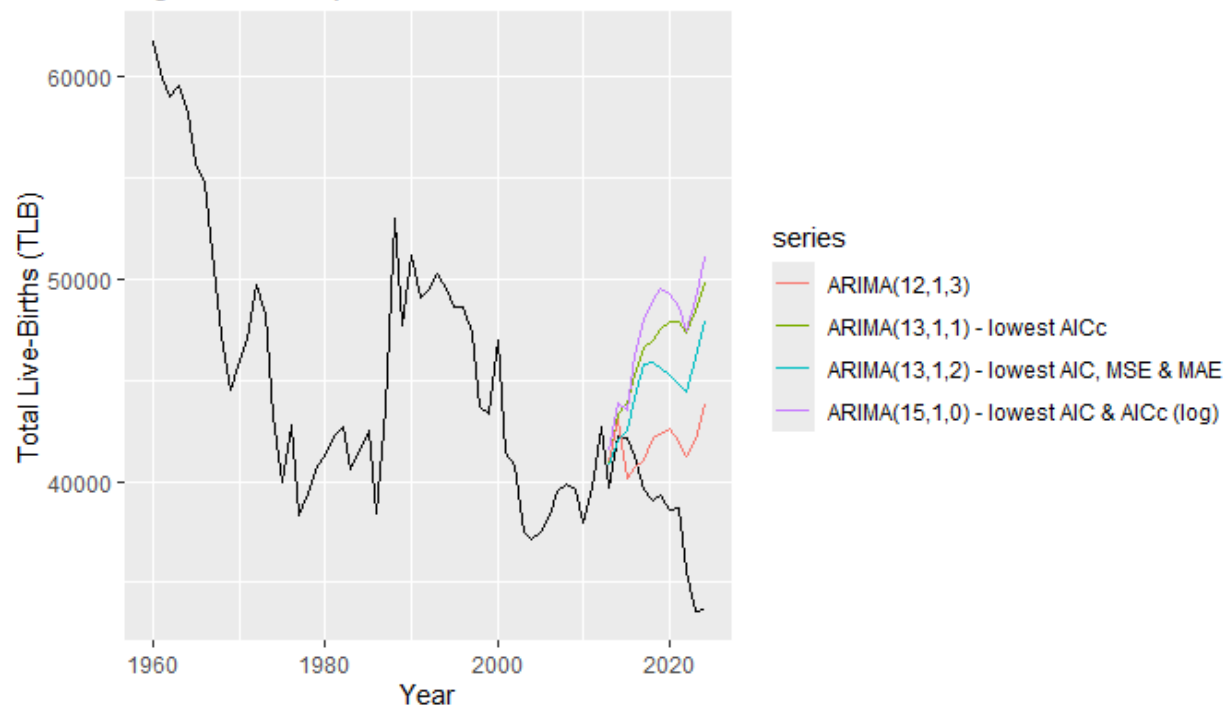
For TLB, the ARIMA(13,1,2) model produced the lowest forecast error and was selected as the preferred model.

Figure 1. Comparison of ARIMA Forecasts for TFR



The forecast plot for TFR indicates that the selected model follows the overall downward trend in the data.

Figure 2. Comparison of ARIMA Forecasts for TLB



The forecast plot for TLB indicates that the candidate models do not fully capture short-term fluctuations in the observed data.

5. Discussion

The results show that model selection solely based on information criteria such as AIC and AICc does not align with forecast accuracy. Across both TFR and TLB, models with lower AIC or AICc values did not consistently produce the lowest forecast errors, highlighting the importance of evaluating models using both in-sample fit and out-of-sample performance.

For TFR, applying a log transformation combined with second-order differencing improved forecast accuracy compared to simpler specifications. This suggests that the series exhibits strong trends and variability that are better captured after transformation and differencing. However, the improved performance is accompanied by increased model complexity, which raises the possibility of overfitting.

For TLB, first-order differencing was sufficient to remove the underlying trend, and logarithmic transformation did not improve forecast performance. This suggests that the variance of TLB is relatively stable, and additional transformations may not meaningfully improve model fit or forecast accuracy.

Despite finding candidate ARIMA models, they are limited by their reliance on historical observations and do not incorporate external variables. This limitation is particularly evident in the absence of an observable increase in the 2024 Dragon year, suggesting that cultural or policy-related effects are not captured by the model structure. Furthermore, several selected models involve relatively high-order autoregressive and moving average terms, increasing model complexity and reinforcing concerns about potential overfitting.

Overall, while ARIMA models are effective at capturing long-term declining trends in TFR and TLB, their ability to account for short-term fluctuations or behavioural changes is limited, which constrains forecast reliability under changing demographic conditions.

6. Conclusion

This report investigates whether ARIMA models can accurately forecast Singapore's TLB and TFR while capturing short-term fluctuations. Overall, the results indicate that ARIMA models are suitable for capturing long-term declining trends in both series, but are less effective at capturing short-term variability and structural fluctuations. Forecast performance varied depending on transformation and differencing choices, with no single model consistently optimal across information criteria and error-based measures. This highlights the importance of balancing model fit with accuracy measures when forecasting demographic time series. The reliance on historical observations limits ARIMA models' ability to incorporate external influences, which in turn may reduce reliability under changing conditions.

8. References

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- World Health Organization. *Total fertility rate (per woman)*. Accessed June 7, 2026, from <https://www.who.int/data/gho/indicator-metadata-registry/imr-details/123>

9. Appendix

Figure 3. Initial Plot of Total Fertility Rate (TFR)

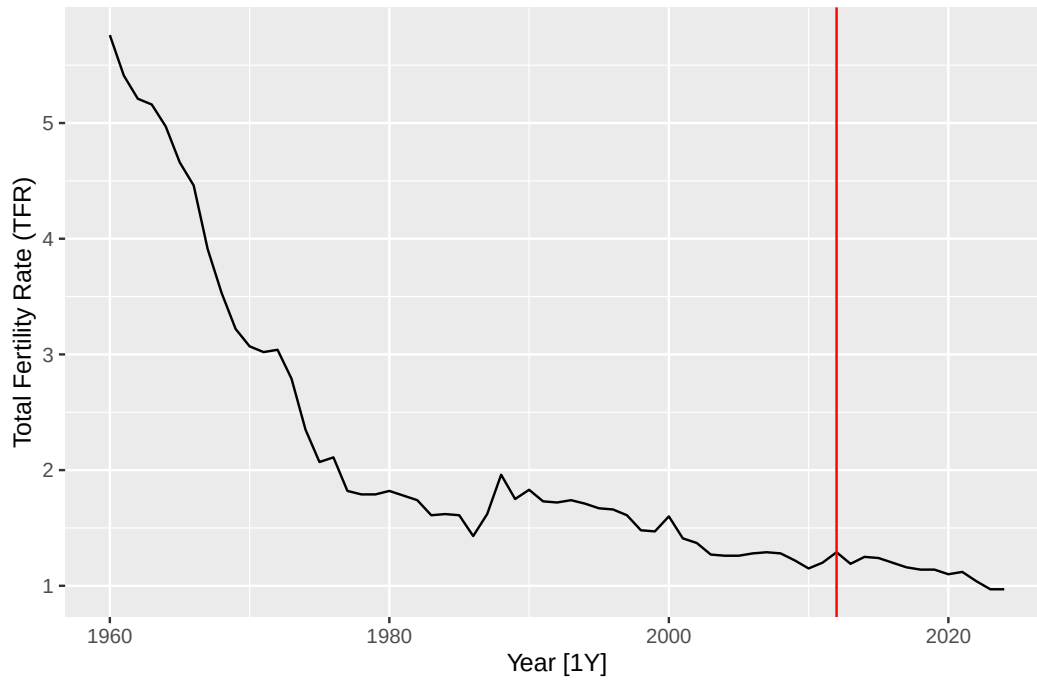
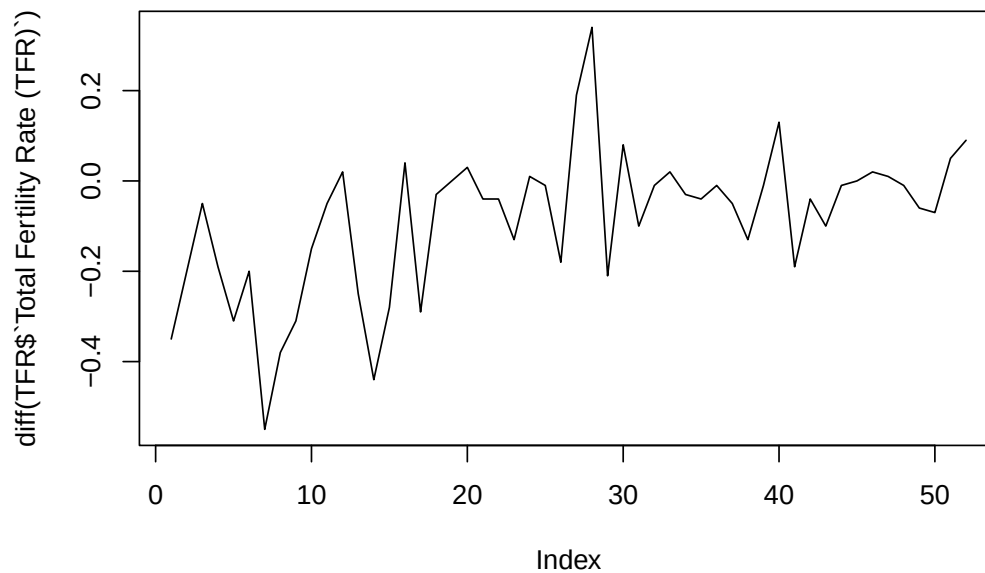
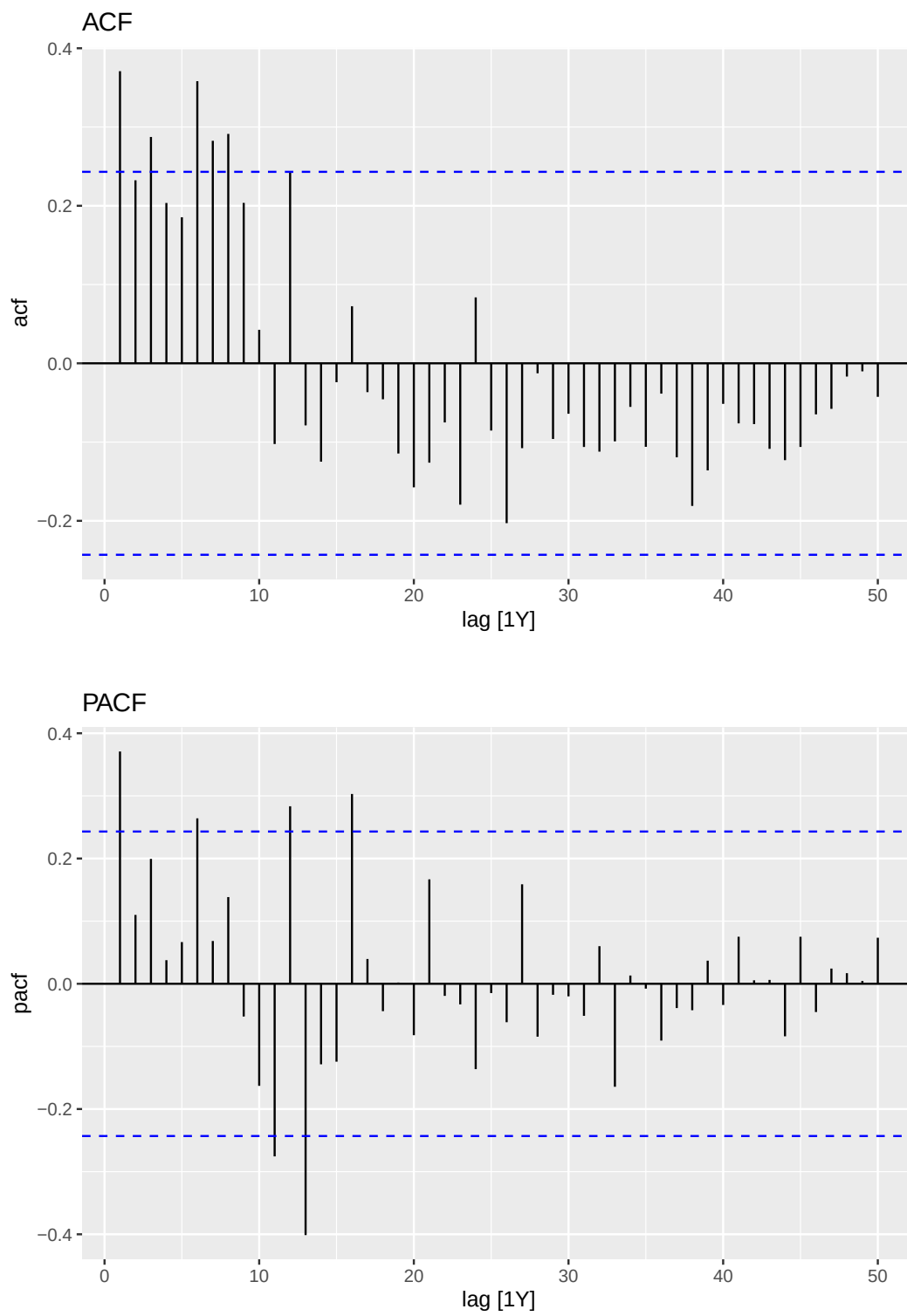


Figure 4. Plot of Differenced TFR until 2012



```
##      kpss_stat kpss_pvalue
##      0.8412374  0.0100000
```

Figure 5. ACF and PACF Plots for Differenced Total Fertility Rate (TFR)




```

## Series: TFR_ts
## ARIMA(16,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.3647 -0.1061  0.4577 -0.2586  0.2984  0.2452  0.2006  0.1578 -0.0293
## s.e.  0.1240  0.1289  0.1246  0.1128  0.0935  0.0832  0.0952  0.0928  0.0996
##      ar10     ar11     ar12     ar13     ar14     ar15     ar16
##      -0.0306 -0.3594  0.3889 -0.6048 -0.0419 -0.3361  0.5135
## s.e.  0.0973  0.0913  0.1039  0.1066  0.1350  0.1295  0.1397
##
## sigma^2 = 0.009716: log likelihood = 48.99
## AIC=-63.99  AICc=-45.99  BIC=-30.82

```

Fitted ARIMA(16,1,0) model for TFR:

$$\begin{aligned}
& (1 - 0.3647B + 0.1061B^2 - 0.4577B^3 + 0.2586B^4 - 0.2984B^5 - 0.2452B^6 \\
& \quad - 0.2006B^7 - 0.1578B^8 + 0.0293B^9 + 0.0306B^{10} + 0.3594B^{11} \\
& \quad - 0.3889B^{12} + 0.6048B^{13} + 0.0419B^{14} + 0.3361B^{15} - 0.5135B^{16})(1 - B)y_t \\
& = c + \varepsilon_t
\end{aligned}$$

Figure 6. ACF and PACF plots of residuals for the ARIMA(16,1,0) model fitted to TFR

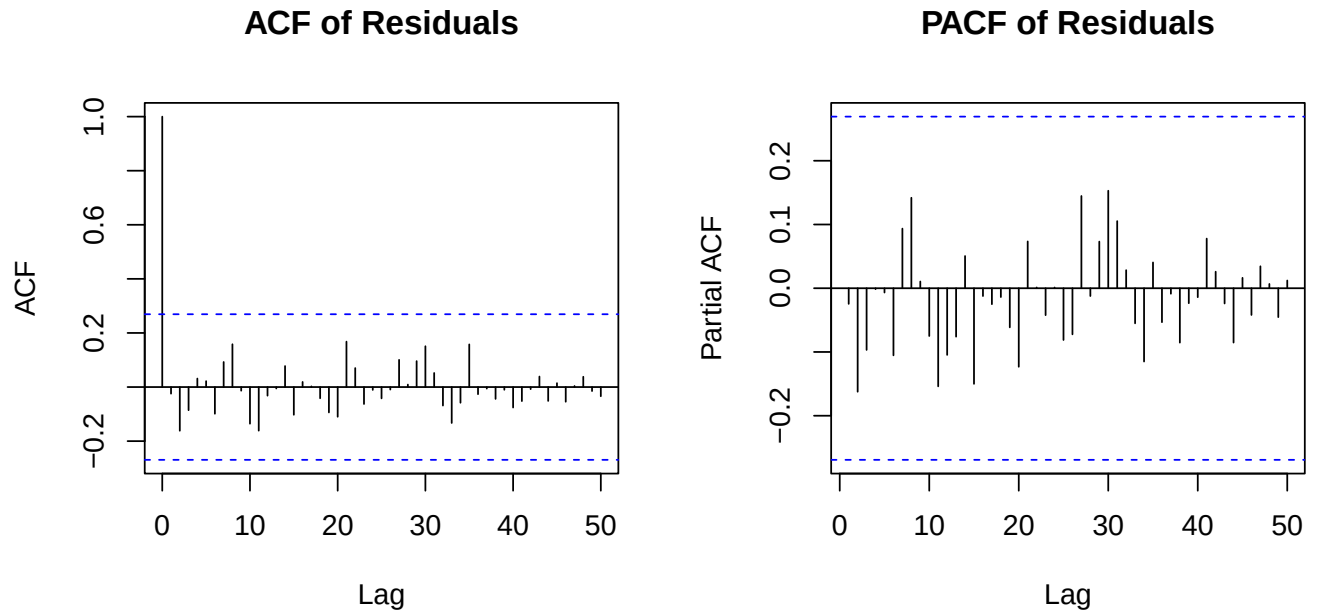
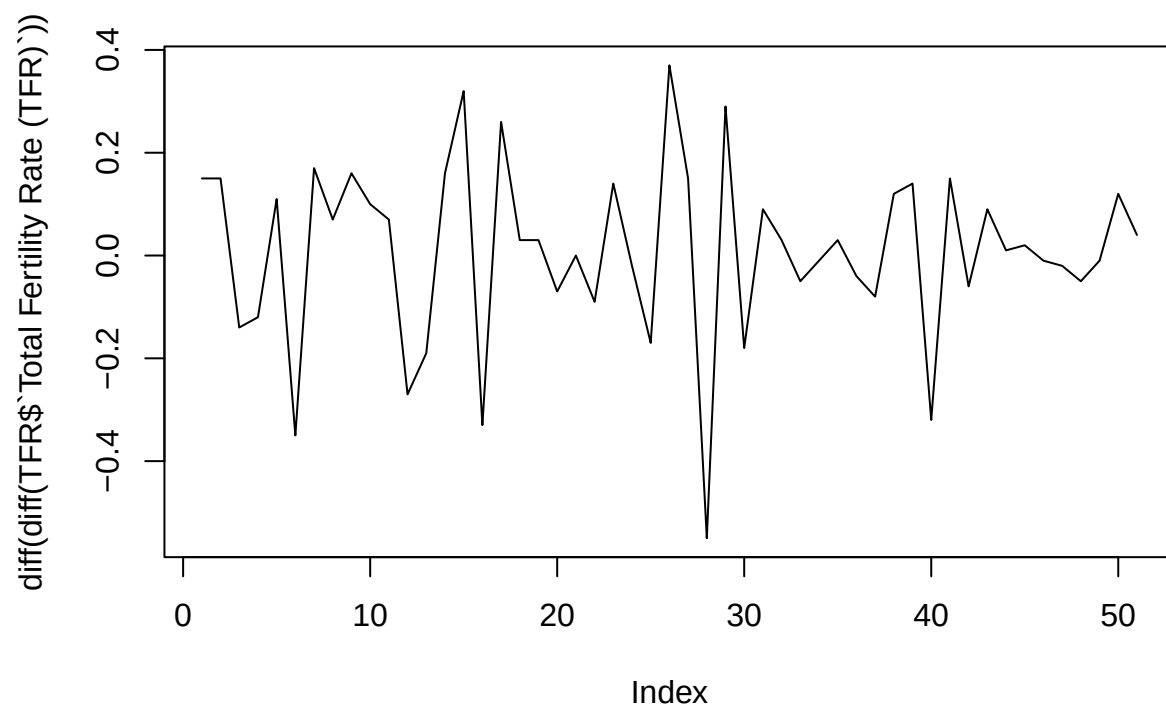
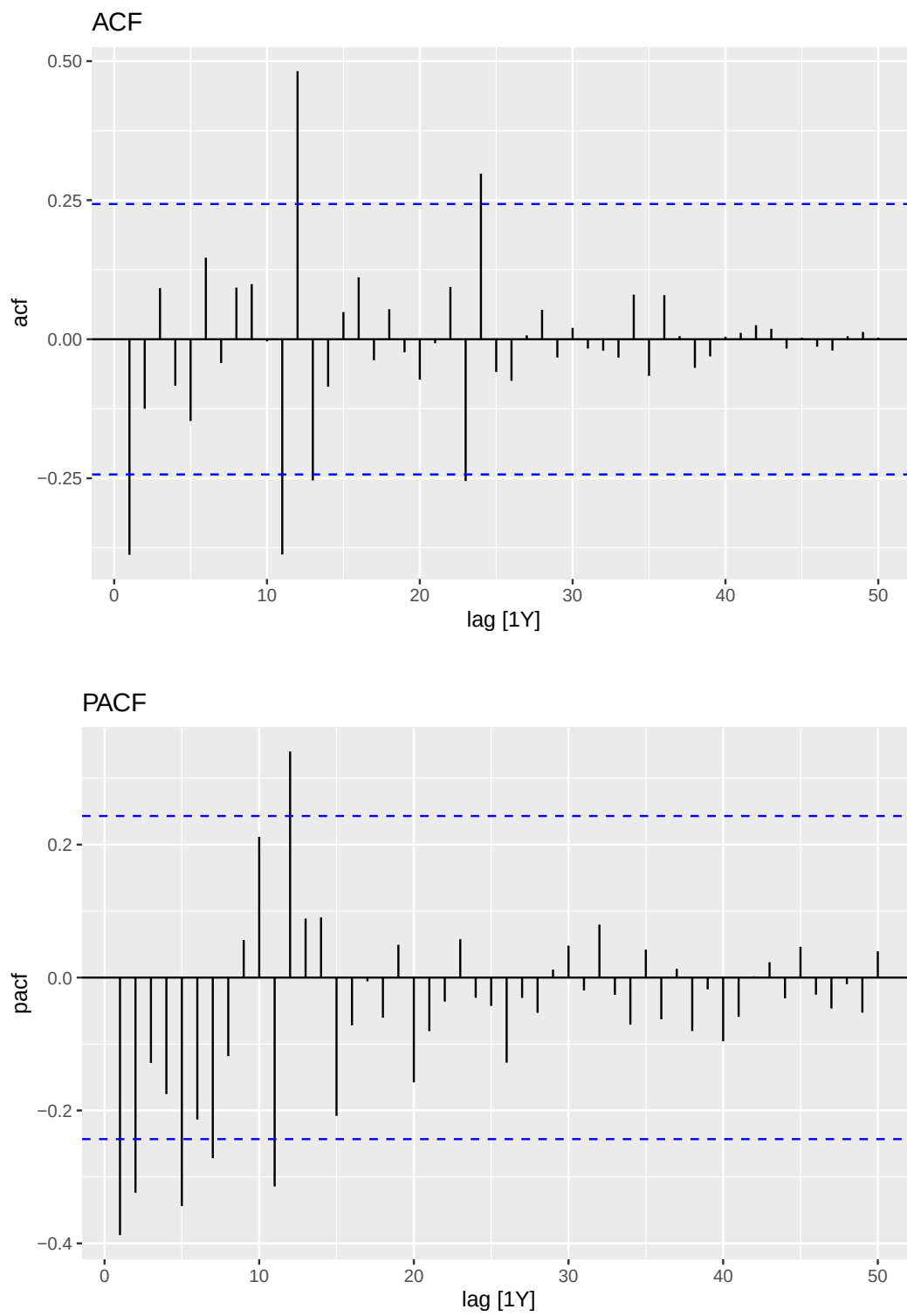


Figure 7. Plot of Differenced Twice TFR until 2012



```
##      kpss_stat kpss_pvalue
## 0.04262877 0.10000000
```

Figure 8. ACF and PACF Plots for Twice Differenced Total Fertility Rate (TFR)

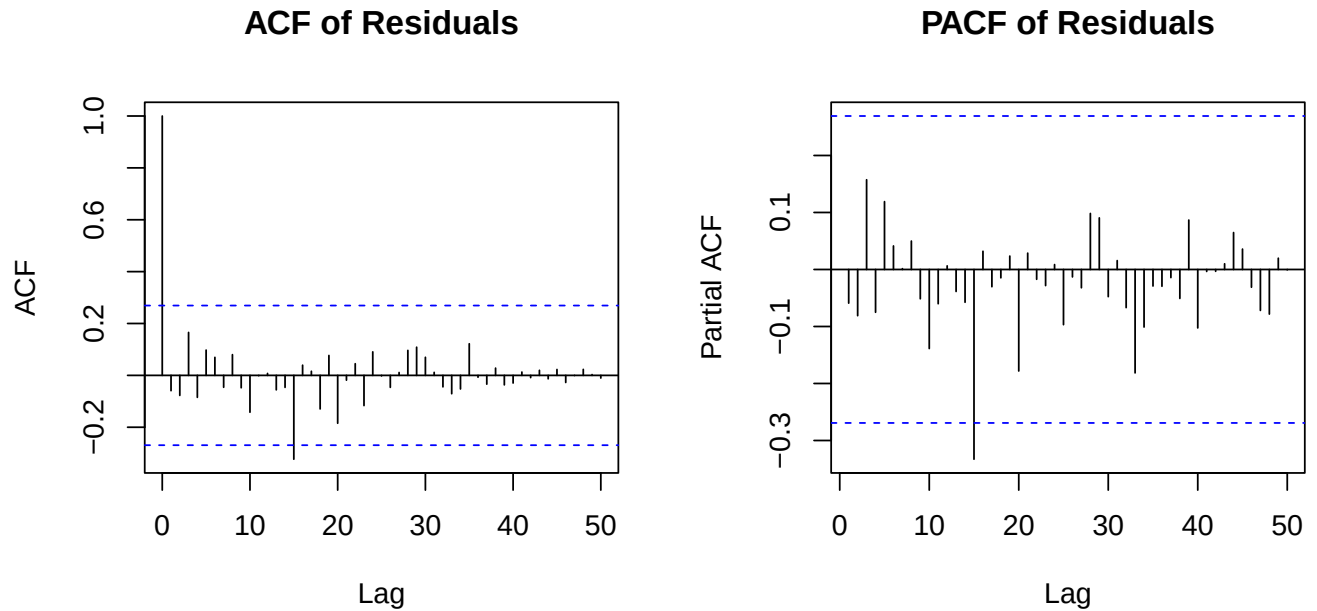


```

## Series: TFR_ts
## ARIMA(12,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.6234 -0.7441 -0.5813 -0.5700 -0.4163 -0.1741  0.0462  0.3482
## s.e.   0.1203   0.1494   0.1673   0.1676   0.1705   0.1772   0.1846   0.1847
##      ar9      ar10     ar11     ar12
##      0.4206   0.4880   0.1609   0.5040
## s.e.   0.1845   0.1783   0.1540   0.1205
##
## sigma^2 = 0.01286:  log likelihood = 41.78
## AIC=-57.55   AICc=-47.72   BIC=-32.44

```

Figure 9. ACF and PACF plots of residuals for the initial model ARIMA(12,2,0) fitted to TFR



```

## Series: TFR_ts
## ARIMA(12,2,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.4622 -0.7606 -0.9273 -0.8808 -0.6967 -0.4124 -0.2312  0.0490
## s.e.  0.1091  0.1266  0.1643  0.2032  0.2376  0.2529  0.2548  0.2404
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1578  0.3268  0.1426  0.6463 -0.1402  0.1411  0.8593
## s.e.  0.2067  0.1645  0.1248  0.1005  0.1237  0.1262  0.1309
##
## sigma^2 = 0.009277: log likelihood = 47.29
## AIC=-62.57  AICc=-46.57  BIC=-31.66

```

Fitted ARIMA(12,2,3) model for TFR:

$$\begin{aligned}
& (1 + 0.4622B + 0.7606B^2 + 0.9273B^3 + 0.8808B^4 + 0.6967B^5 + 0.4124B^6 \\
& \quad + 0.2312B^7 - 0.0490B^8 - 0.1578B^9 - 0.3268B^{10} - 0.1426B^{11} - 0.6463B^{12})(1 - B)^2 y_t \\
& = (1 - 0.1402B + 0.1411B^2 + 0.8593B^3) \varepsilon_t + c
\end{aligned}$$

Figure 10. ACF and PACF plots of residuals for the ARIMA(12,2,3) model fitted to TFR

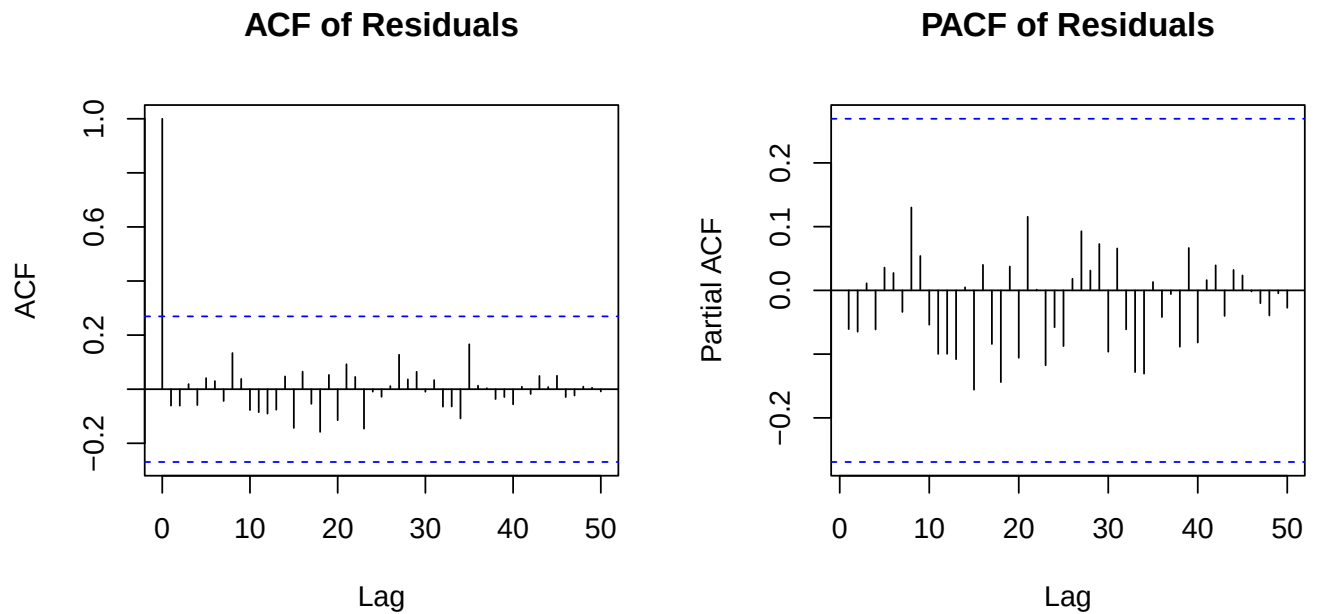


Figure 11. Plot of Log Transformation for TFR until 2012

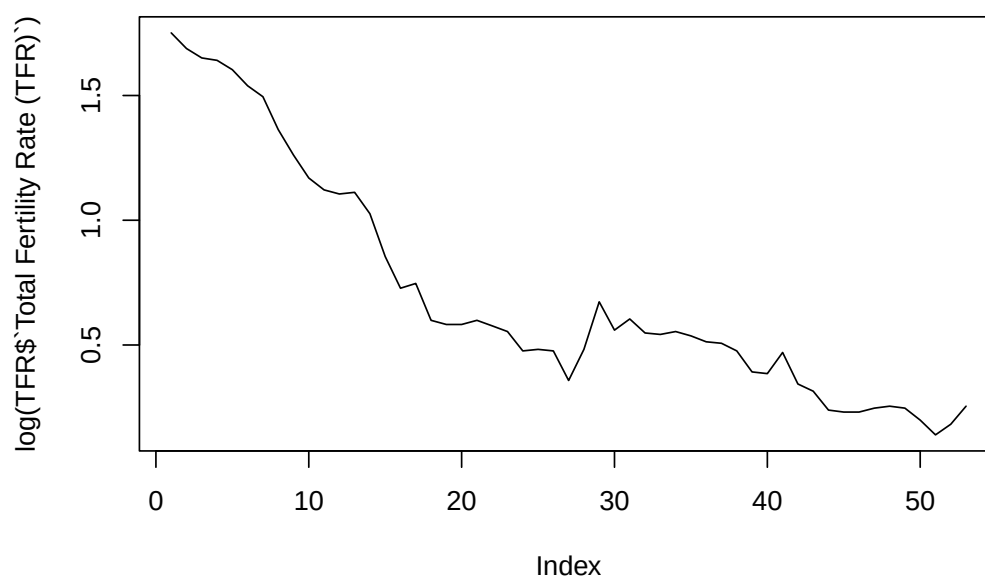
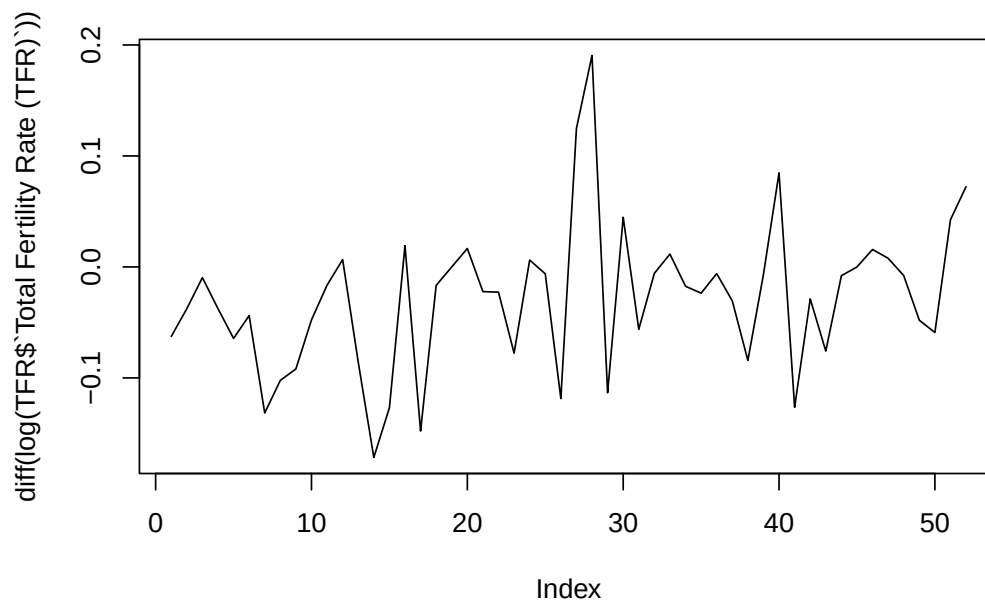
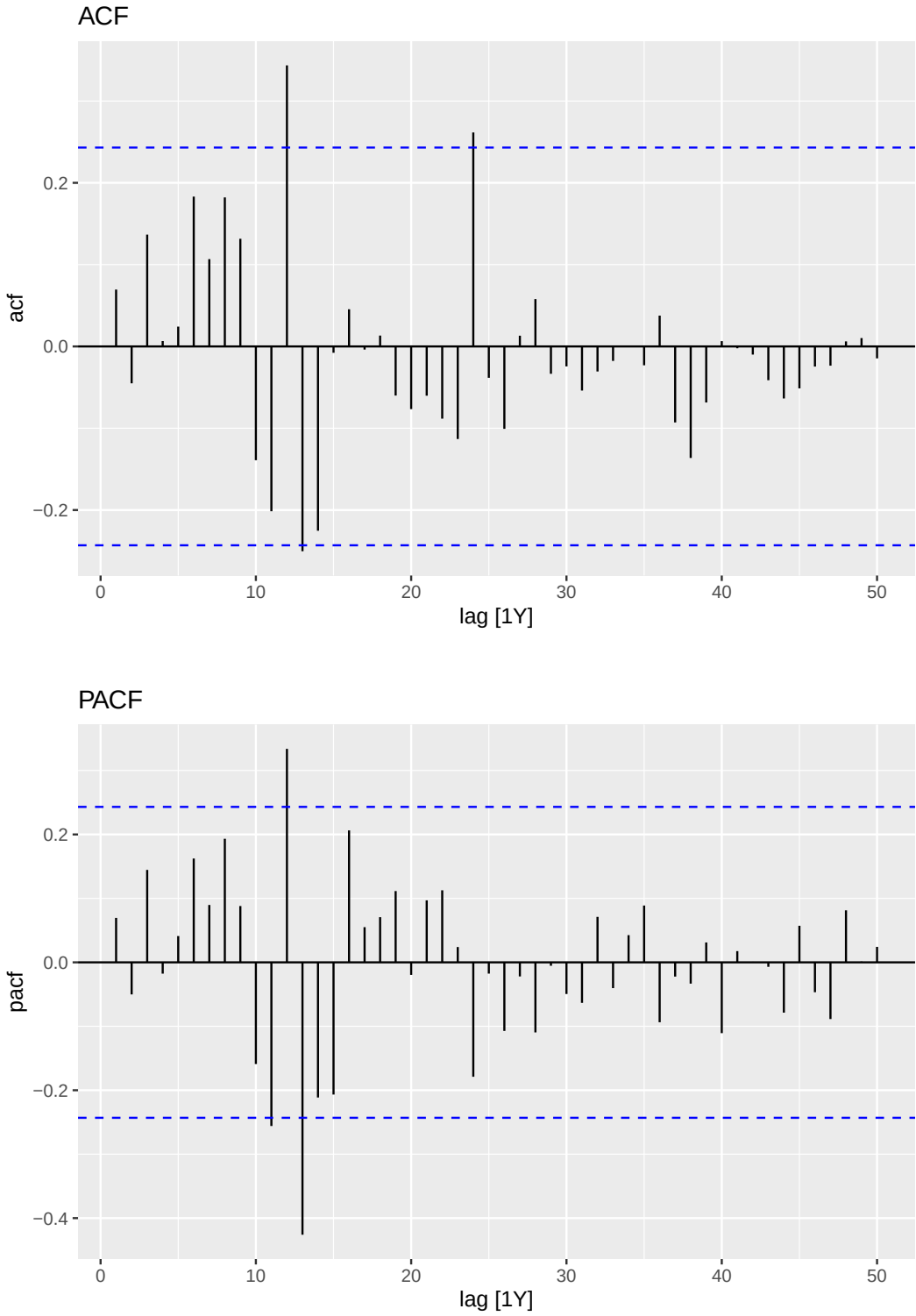


Figure 12. Plot of Differenced Log Transformation for TFR until 2012



```
##      kpss_stat kpss_pvalue
## 0.50484854 0.04057465
```

Figure 13. ACF and PACF Plots for Differenced Log Transformation of Total Fertility Rate (TFR)

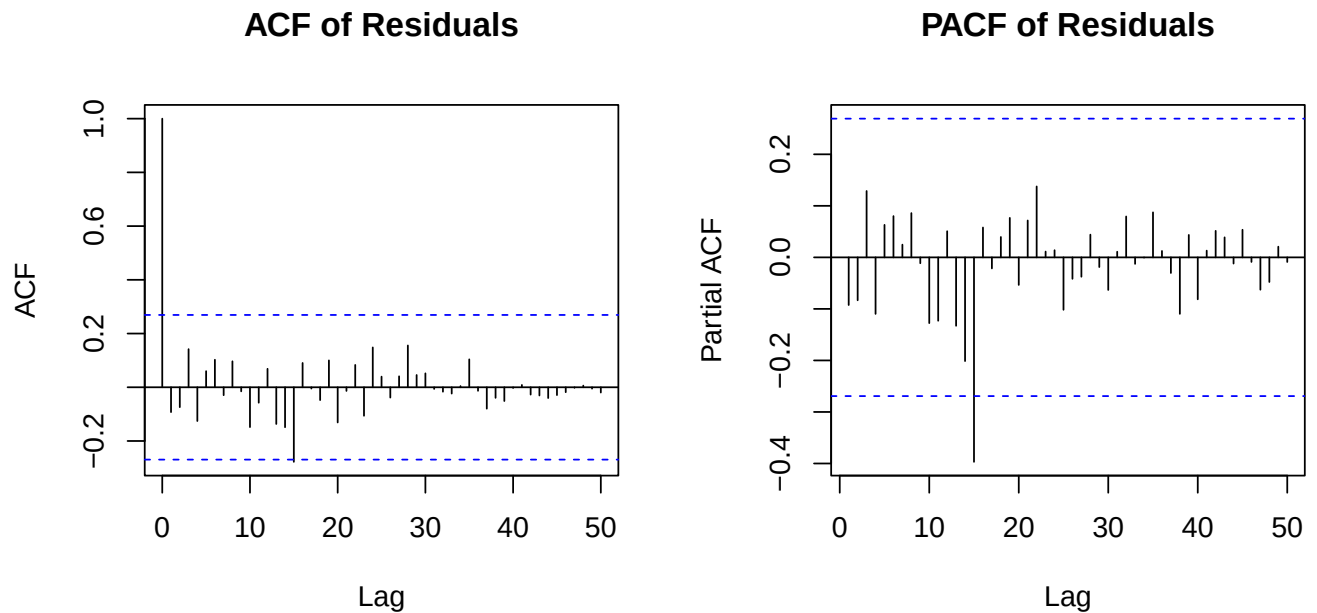


```

## Series: log(TFR_ts)
## ARIMA(13,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2629 -0.0799  0.1194  0.0024  0.1349  0.1715  0.1679  0.2392  0.0298
## s.e.  0.1215  0.1145  0.1065  0.1064  0.1079  0.0974  0.1023  0.1036  0.1106
##      ar10     ar11     ar12     ar13
##      -0.0307 -0.2349  0.3587 -0.4375
## s.e.   0.1130   0.1084   0.1170   0.1258
##
## sigma^2 = 0.00323: log likelihood = 79.9
## AIC=-131.8  AICc=-120.45  BIC=-104.48

```

Figure 14. ACF and PACF plots of residuals for the initial log transformed model: ARIMA(13,1,0) model fitted to TFR




```

## Series: log(TFR_ts)
## ARIMA(13,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2783 -0.2434 -0.2961 -0.0013  0.1400  0.2218  0.1567  0.2497  0.0839
## s.e.  0.1256  0.1019  0.1096  0.1094  0.1091  0.0975  0.1024  0.1087  0.1147
##      ar10     ar11     ar12     ar13     ma1      ma2      ma3
##      0.0722 -0.0719  0.4674 -0.5235  0.0903  0.1376  0.9749
## s.e.  0.1170  0.1097  0.1005  0.1256  0.0271  0.1085  0.1089
##
## sigma^2 = 0.00232:  log likelihood = 85
## AIC=-136   AICc=-118   BIC=-102.83

```

Fitted ARIMA(13,1,3) model for TFR:

$$\begin{aligned}
& (1 - 0.2783B + 0.2434B^2 + 0.2961B^3 + 0.0013B^4 - 0.1400B^5 - 0.2218B^6 \\
& \quad - 0.1567B^7 - 0.2497B^8 - 0.0839B^9 - 0.0722B^{10} + 0.0719B^{11} - 0.4674B^{12} + 0.5235B^{13})(1 - B) \log(y_t) \\
& = (1 - 0.0903B - 0.1376B^2 - 0.9749B^3)\varepsilon_t + c
\end{aligned}$$

Figure 15. ACF and PACF plots of residuals for the ARIMA(13,1,3) model fitted to TFR with log transformation

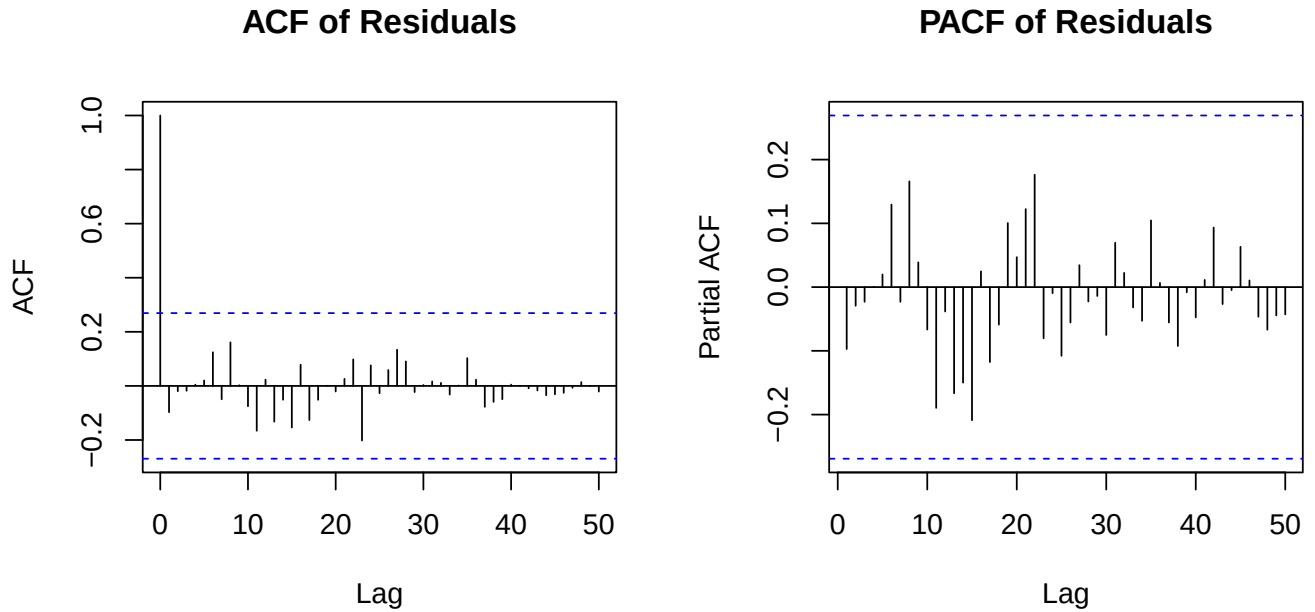
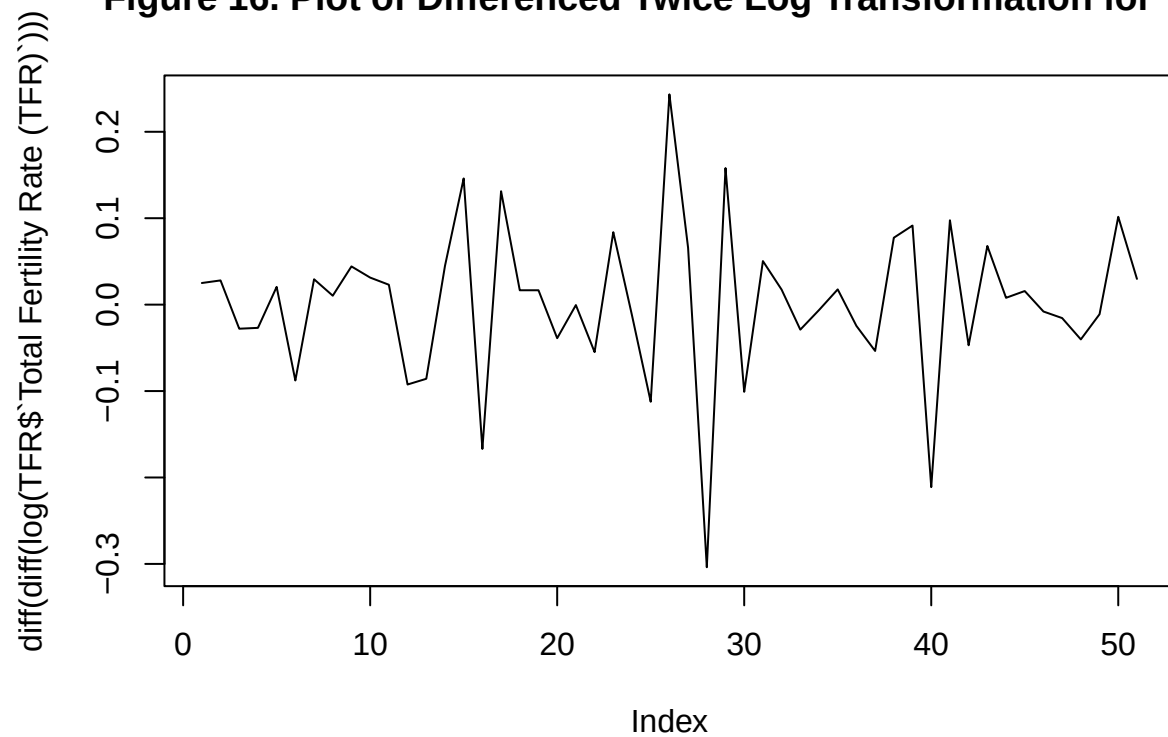
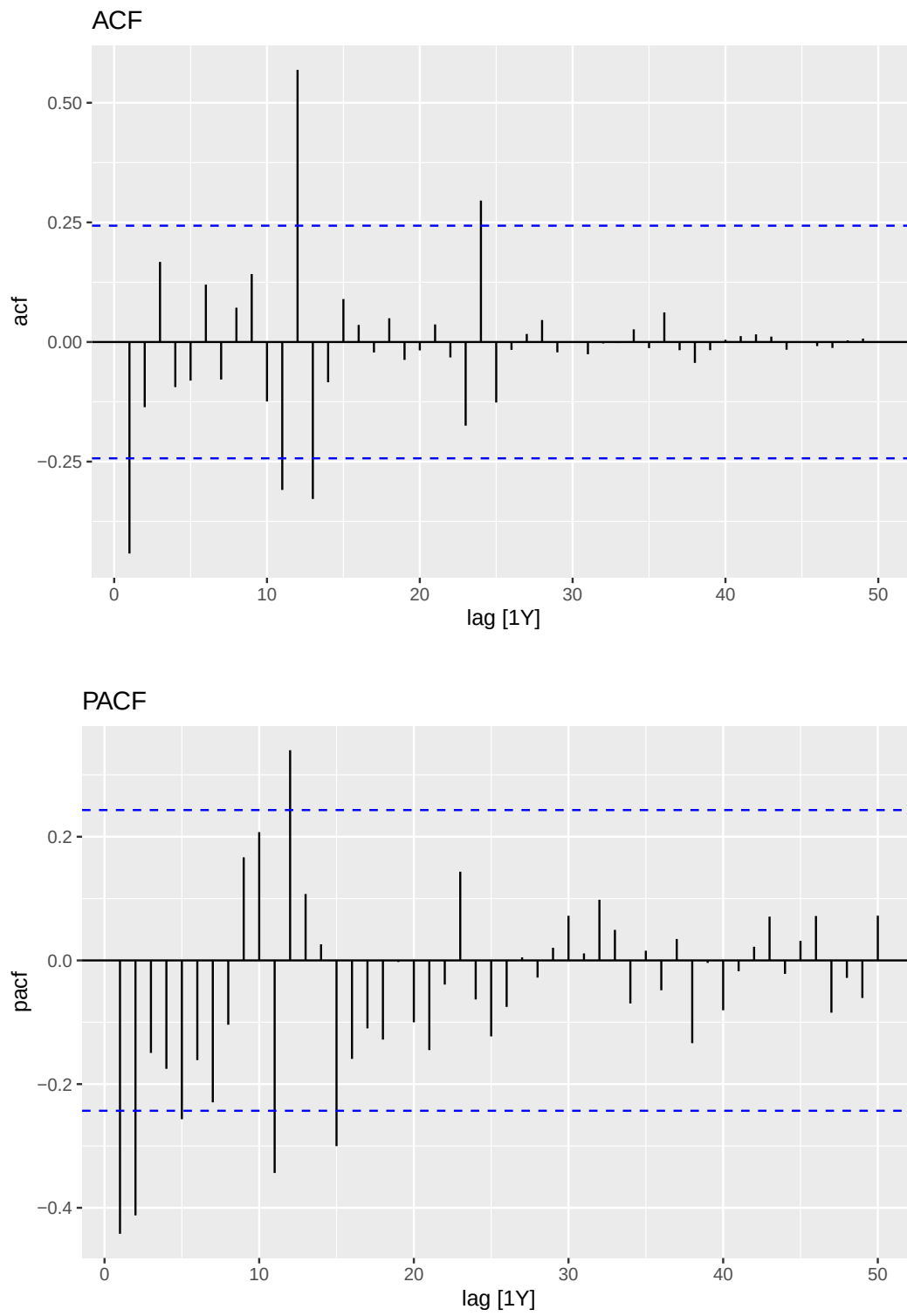


Figure 16. Plot of Differenced Twice Log Transformation for TFR



```
##      kpss_stat kpss_pvalue
## 0.04760604 0.10000000
```

Figure 17. ACF and PACF Plots for Twice Differenced Log Transformation of TFR

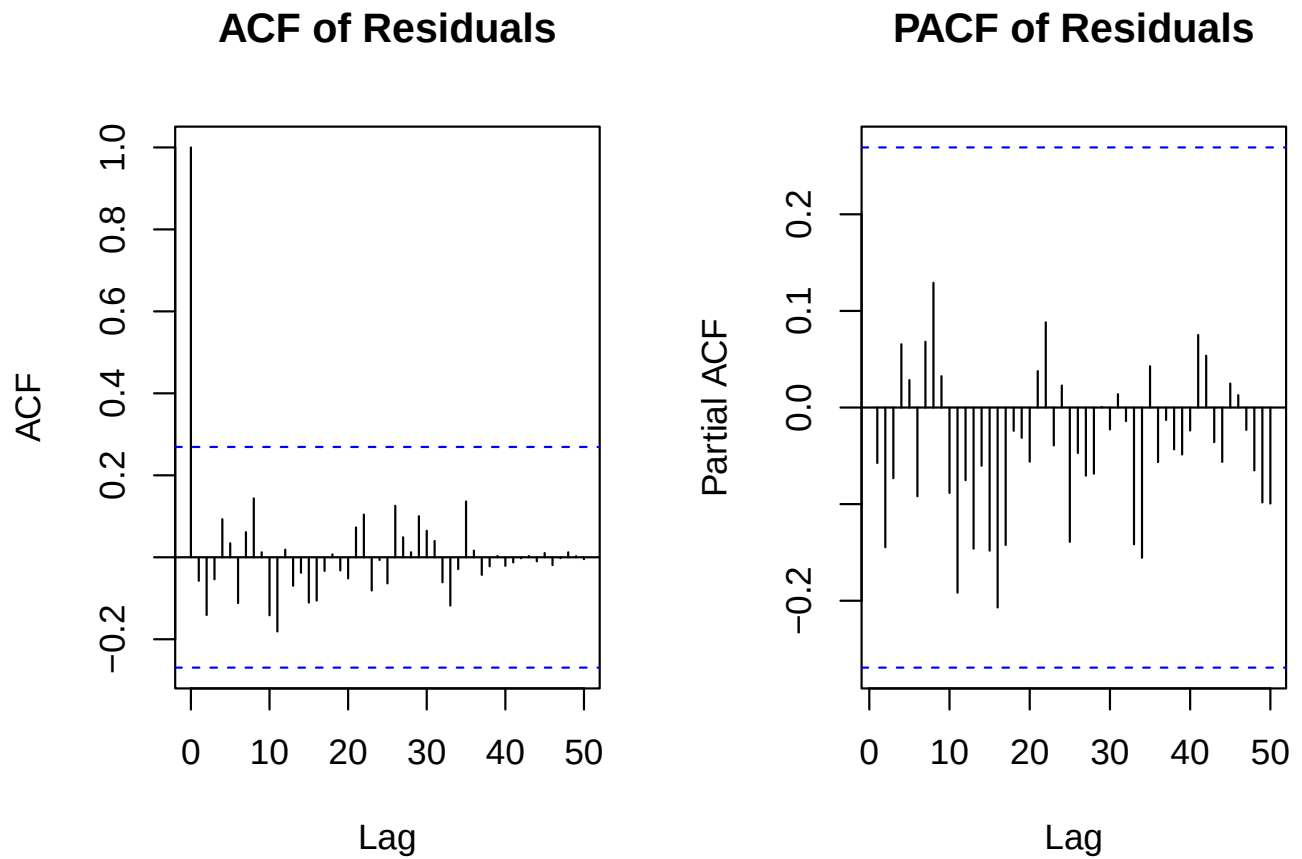


```

## Series: log(TFR_ts)
## ARIMA(15,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.7137 -0.7004 -0.3372 -0.5826 -0.3331 -0.1103  0.0628  0.2087
## s.e.   0.1218  0.1571  0.1913  0.1824  0.2031  0.2111  0.2146  0.2149
##      ar9      ar10     ar11     ar12     ar13     ar14     ar15
##     0.1864  0.1142  -0.1162  0.3260  -0.1661  -0.2596  -0.4876
## s.e.   0.2152  0.2137   0.2042  0.1855   0.1901   0.1618   0.1289
##
## sigma^2 = 0.002602: log likelihood = 82.83
## AIC=-133.67  AICc=-117.67  BIC=-102.76

```

Figure 18. ACF and PACF plots of residuals for the ARIMA(15,2,0) model fitted to TFR with log transformation



```

## Series: log(TFR_ts)
## ARIMA(13,2,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6514 -0.2412 -0.0063 -0.1608  0.0605  0.0715  0.0280  0.1159
## s.e.  0.1194  0.1202  0.1193  0.1180  0.1158  0.1095  0.1099  0.1125
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.1105 -0.0649 -0.1966  0.4240 -0.6274 -1.6346  0.7851
## s.e.  0.1156  0.1159  0.1124  0.1126  0.0987  0.1446  0.1392
##
## sigma^2 = 0.002717: log likelihood = 82.19
## AIC=-132.39  AICc=-116.39  BIC=-101.48

```

Fitted ARIMA(13,2,2) model for TFR:

$$\begin{aligned}
& (1 - 0.6514B + 0.2412B^2 + 0.0063B^3 + 0.1608B^4 - 0.0605B^5 - 0.0715B^6 \\
& \quad - 0.0280B^7 - 0.1159B^8 + 0.1105B^9 + 0.0649B^{10} + 0.1966B^{11} - 0.4240B^{12} + 0.6274B^{13})(1 - B)^2 \log(y_t) \\
& = (1 + 1.6346B - 0.7851B^2)\varepsilon_t + c
\end{aligned}$$

Figure 19. ACF and PACF plots of residuals for the ARIMA(13,2,2) model fitted to TFR with log transformation

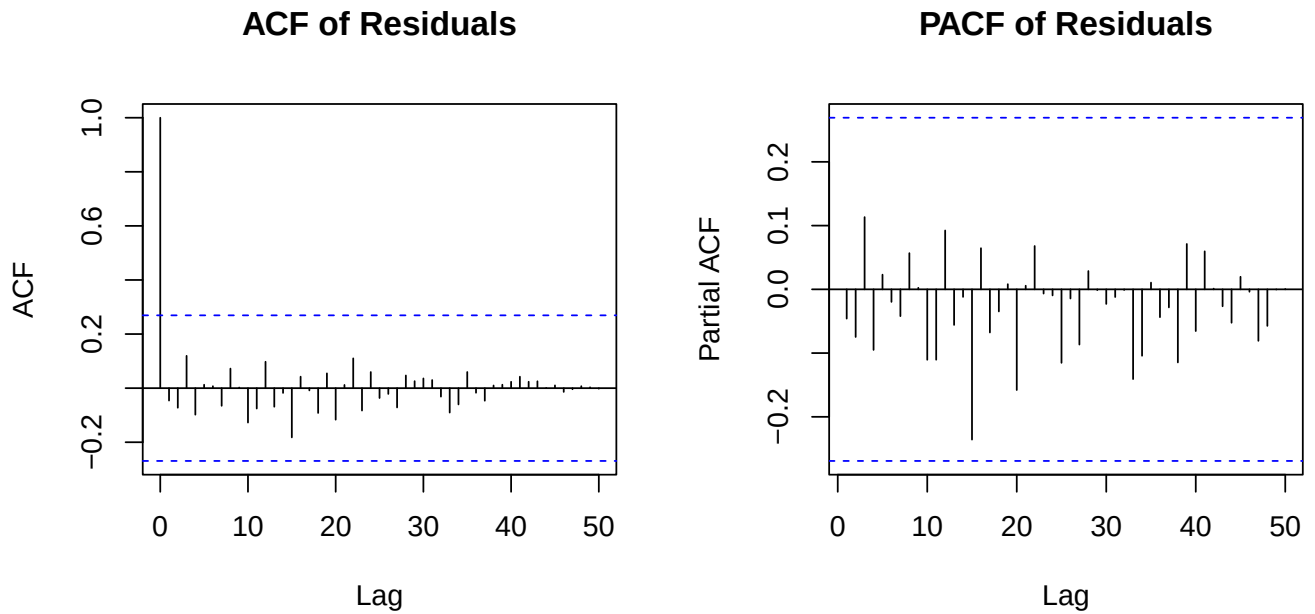


Figure 20. Initial Plot of Total Live-Births

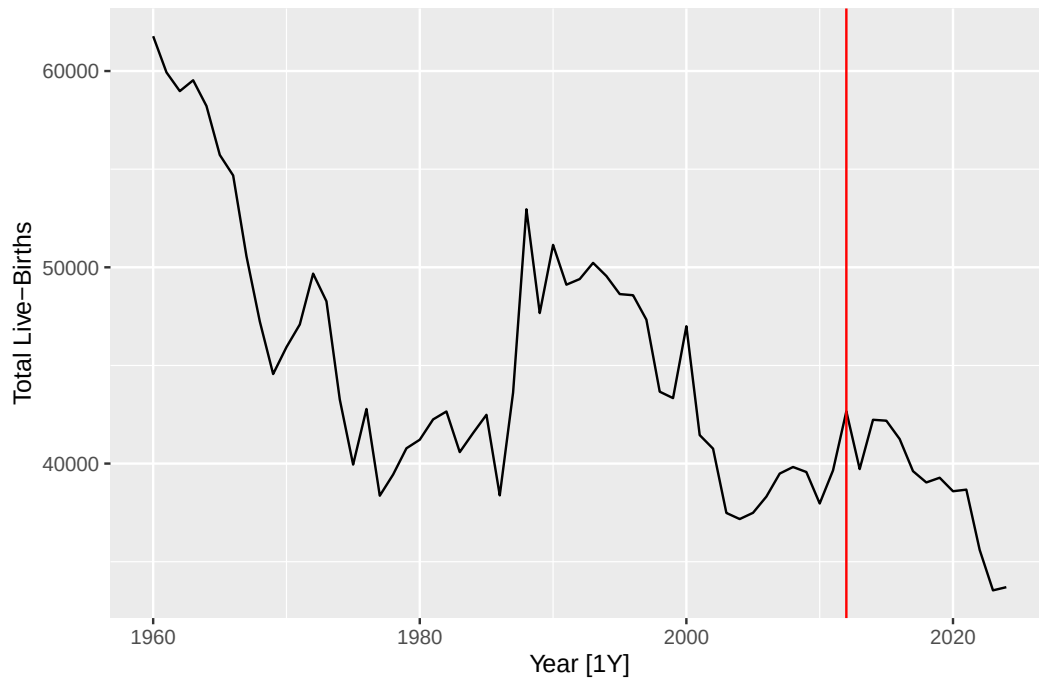
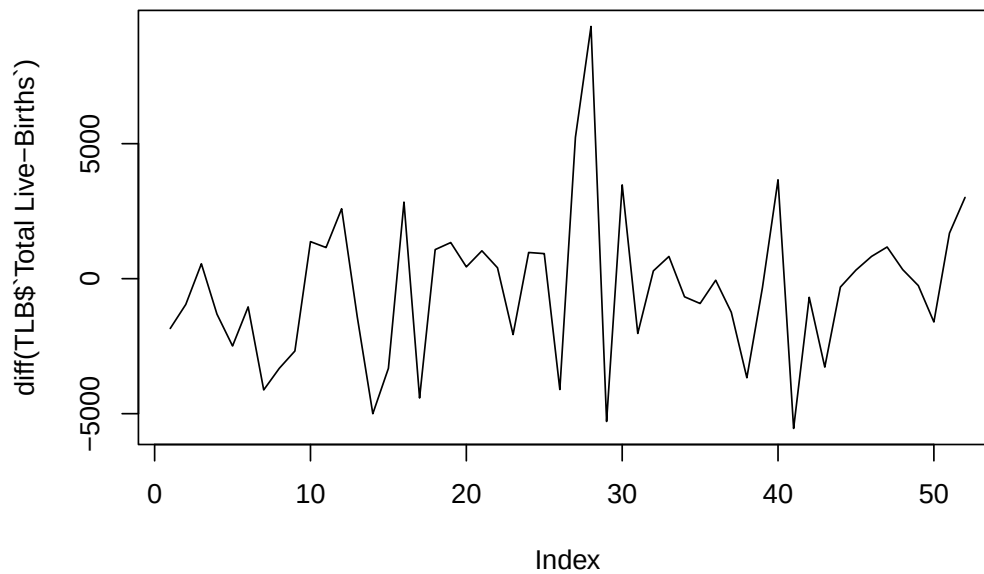
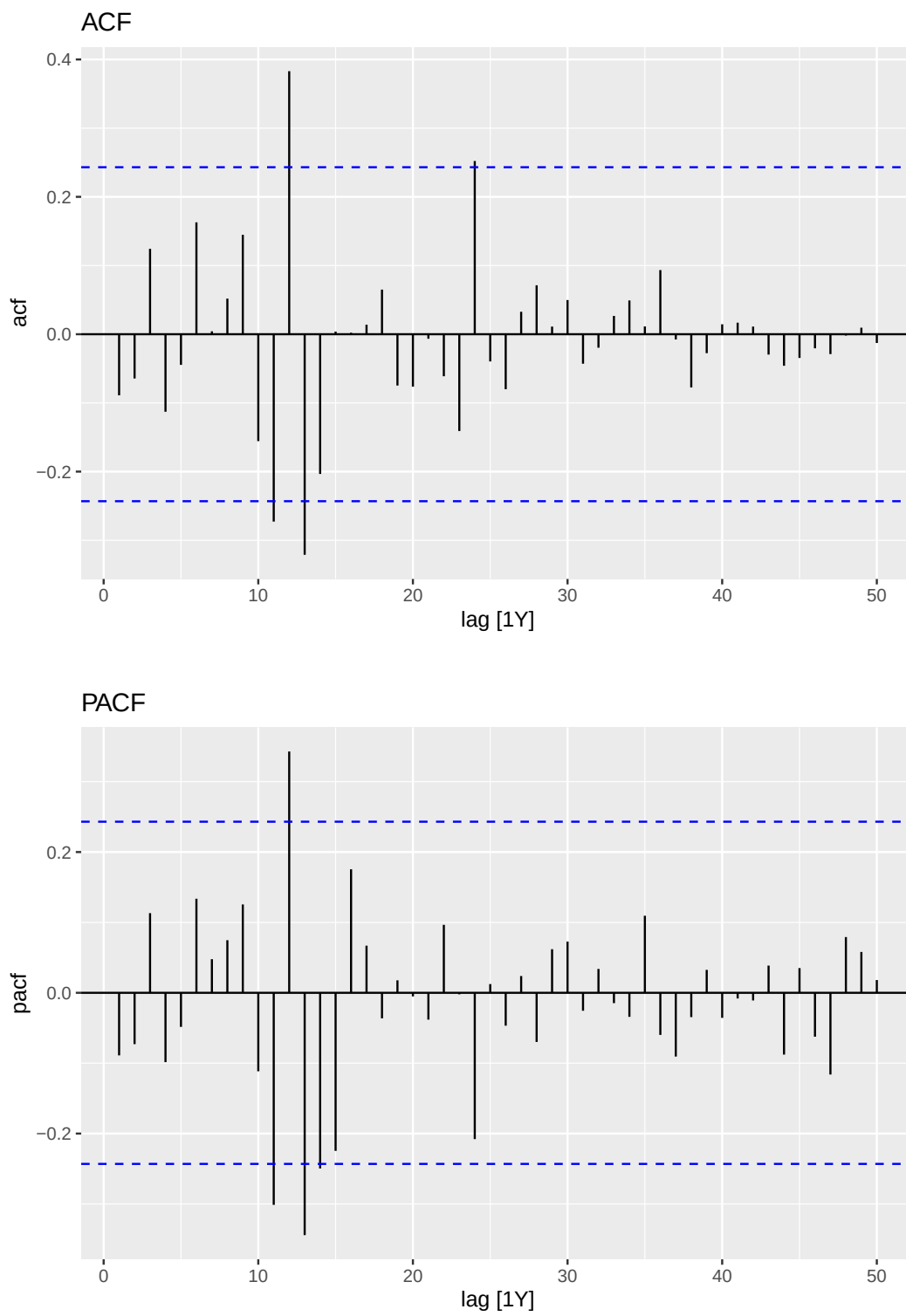


Figure 21. Plot of Differenced Total Live Births (TLB) until 2012



```
##      kpss_stat kpss_pvalue
## 0.2126452    0.1000000
```

Figure 22. ACF and PACF Plots for Differenced Total Live Births (TLB)



```

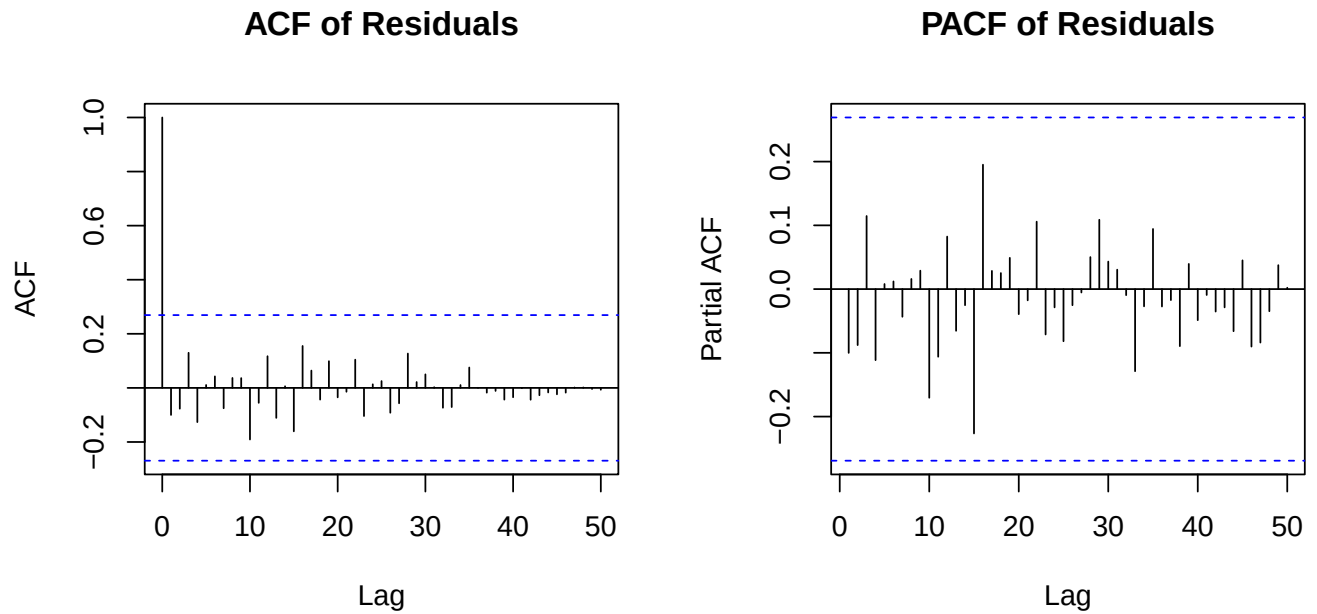
## Series: TLB_ts
## ARIMA(14,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.0260  0.0380 -0.0184 -0.0555  0.0794  0.1459  0.1111  0.1666  0.0608
## s.e.  0.1344  0.1218  0.1091  0.1027  0.1050  0.1015  0.1087  0.1051  0.1087
##      ar10     ar11     ar12     ar13     ar14
##      -0.0696 -0.2211  0.2823 -0.3715 -0.2759
## s.e.   0.1085   0.1078  0.1195   0.1245   0.1466
##
## sigma^2 = 5629775:  log likelihood = -472.66
## AIC=975.33   AICc=988.66   BIC=1004.6

```

Fitted ARIMA(14,1,0) model for TLB:

$$\begin{aligned}
& (1 - 0.0260B - 0.0380B^2 + 0.0184B^3 + 0.0555B^4 - 0.0794B^5 - 0.1459B^6 \\
& \quad - 0.1111B^7 - 0.1666B^8 - 0.0608B^9 + 0.0696B^{10} + 0.2211B^{11} - 0.2823B^{12} + 0.3715B^{13} + 0.2759B^{14})(1 - B)y_t \\
& = c + \varepsilon_t
\end{aligned}$$

Figure 23. ACF and PACF plots of residuals for ARIMA(14,1,0) model fitted to TLB




```

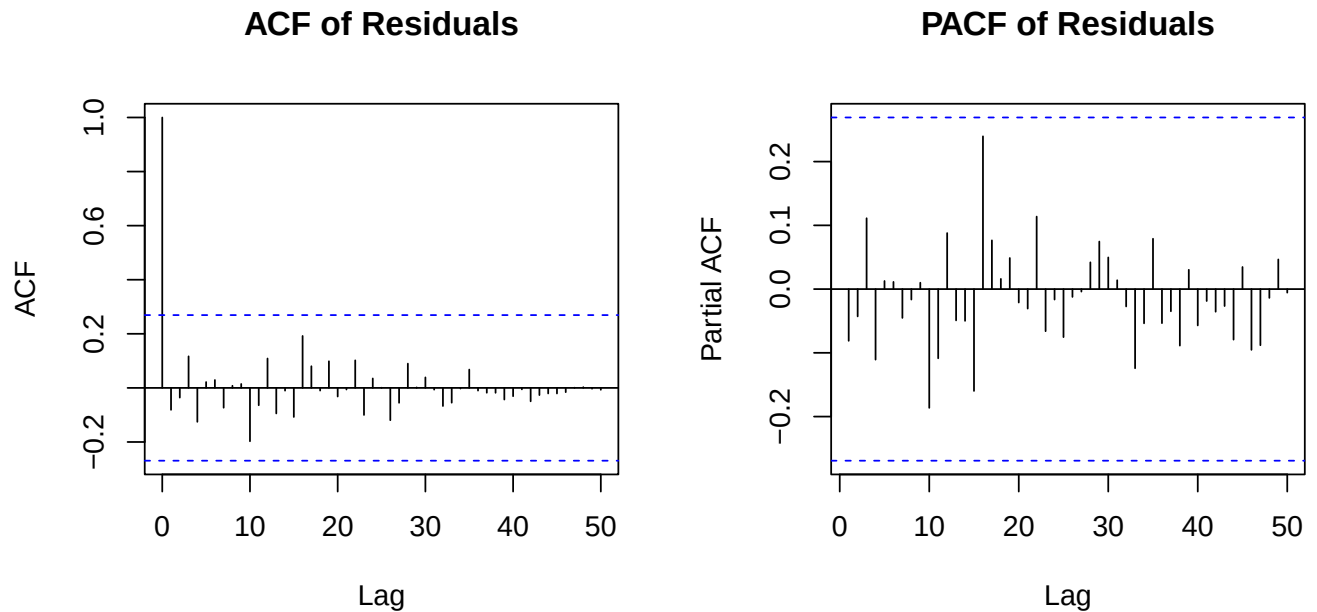
## Series: TLB_ts
## ARIMA(13,1,1)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.5571 -0.0703  0.0264 -0.0762  0.0940  0.1004  0.0542  0.1077 -0.0216
## s.e.  0.1677  0.1156  0.1116  0.1120  0.1104  0.1040  0.1090  0.1095  0.1125
##      ar10     ar11     ar12     ar13     ma1
##      -0.0853 -0.1964  0.4091 -0.5605 -0.5344
## s.e.   0.1130  0.1112  0.1223  0.1126  0.1886
##
## sigma^2 = 5503186: log likelihood = -472.28
## AIC=974.57  AICc=987.9  BIC=1003.83

```

Fitted ARIMA(13,1,1) model for TLB:

$$\begin{aligned}
& (1 - 0.5571B + 0.0703B^2 - 0.0264B^3 + 0.0762B^4 - 0.0940B^5 - 0.1004B^6 \\
& - 0.0542B^7 - 0.1077B^8 + 0.0216B^9 + 0.0853B^{10} + 0.1964B^{11} - 0.4091B^{12} + 0.5605B^{13})(1 - B)y_t \\
& = (1 + 0.5344B)\varepsilon_t + c
\end{aligned}$$

Figure 24. ACF and PACF plots of residuals for ARIMA(13,1,1) model fitted to TLB



```

## Series: TLB_ts
## ARIMA(13,1,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6648 -0.4027 -0.0077 -0.0835  0.1408  0.0861  0.0430  0.1468
## s.e.  0.1376  0.1968  0.1260  0.1252  0.1252  0.1203  0.1199  0.1246
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.0053 -0.0478 -0.1305  0.4076 -0.6385 -0.7326  0.5855
## s.e.  0.1262  0.1271  0.1253  0.1159  0.1041  0.1611  0.3171
##
## sigma^2 = 5122169: log likelihood = -470.51
## AIC=973.01  AICc=988.56  BIC=1004.23

```

Fitted ARIMA(13,1,2) model for TLB:

$$\begin{aligned}
& (1 - 0.6648B + 0.4027B^2 + 0.0077B^3 + 0.0835B^4 - 0.1408B^5 - 0.0861B^6 \\
& \quad - 0.0430B^7 - 0.1468B^8 + 0.0053B^9 + 0.0478B^{10} + 0.1305B^{11} - 0.4076B^{12} + 0.6385B^{13})(1 - B)y_t \\
& = (1 + 0.7326B - 0.5855B^2)\varepsilon_t + c
\end{aligned}$$

Figure 25. ACF and PACF plots of residuals for ARIMA(13,1,2) model fitted to TLB

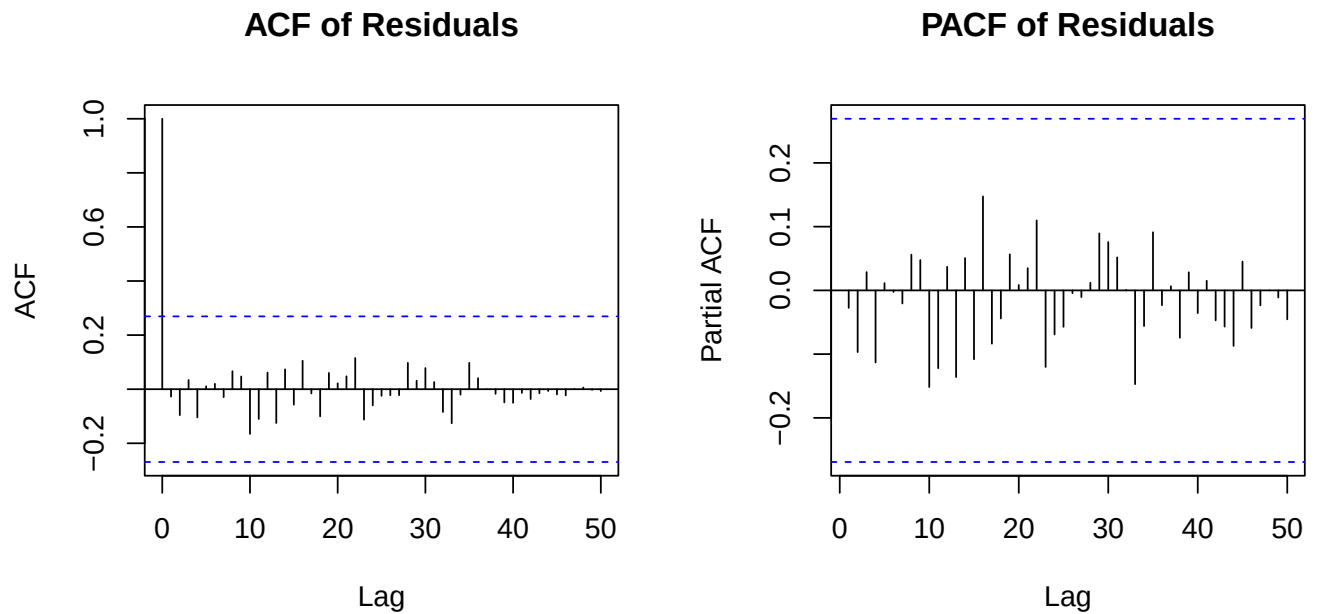


Figure 26. Plot of Log Transformation for TLB until 2012

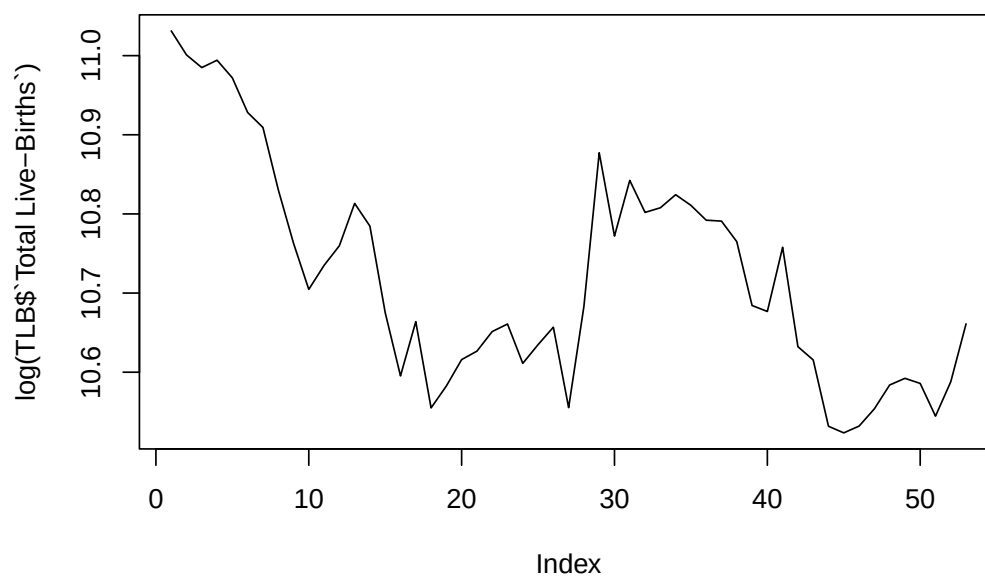
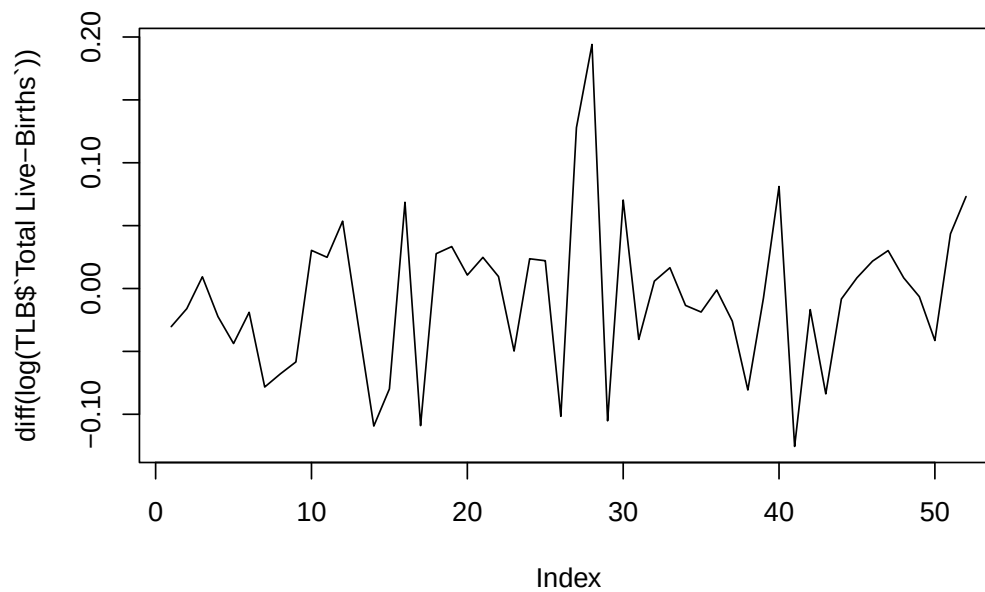
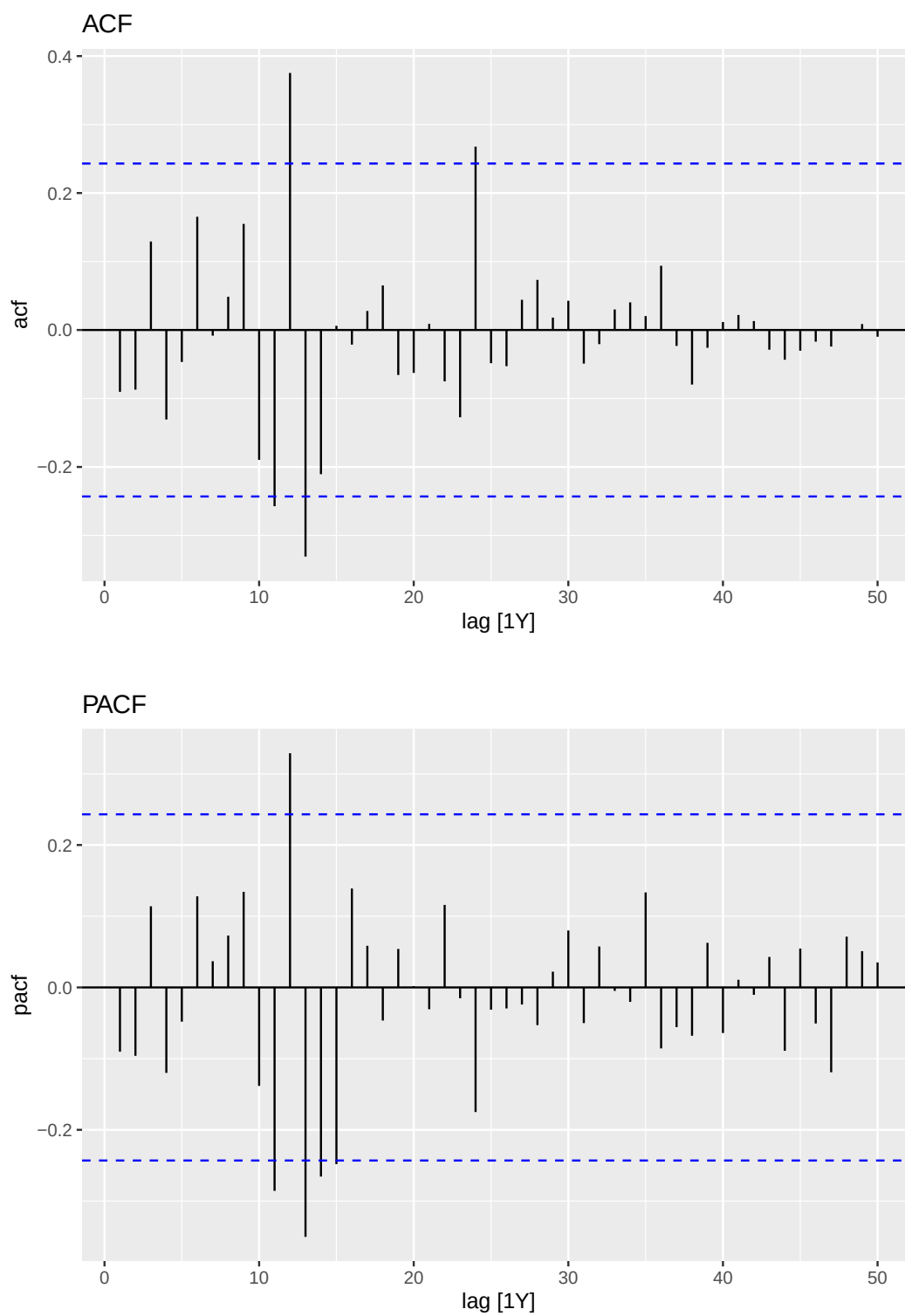


Figure 27. Plot of Differenced Log Transformation for TLB until 2012



```
##      kpss_stat kpss_pvalue
##      0.1765042  0.1000000
```

Figure 28. ACF and PACF Plot for Differenced Log Transformation of Total Live-Births (TLB)



```

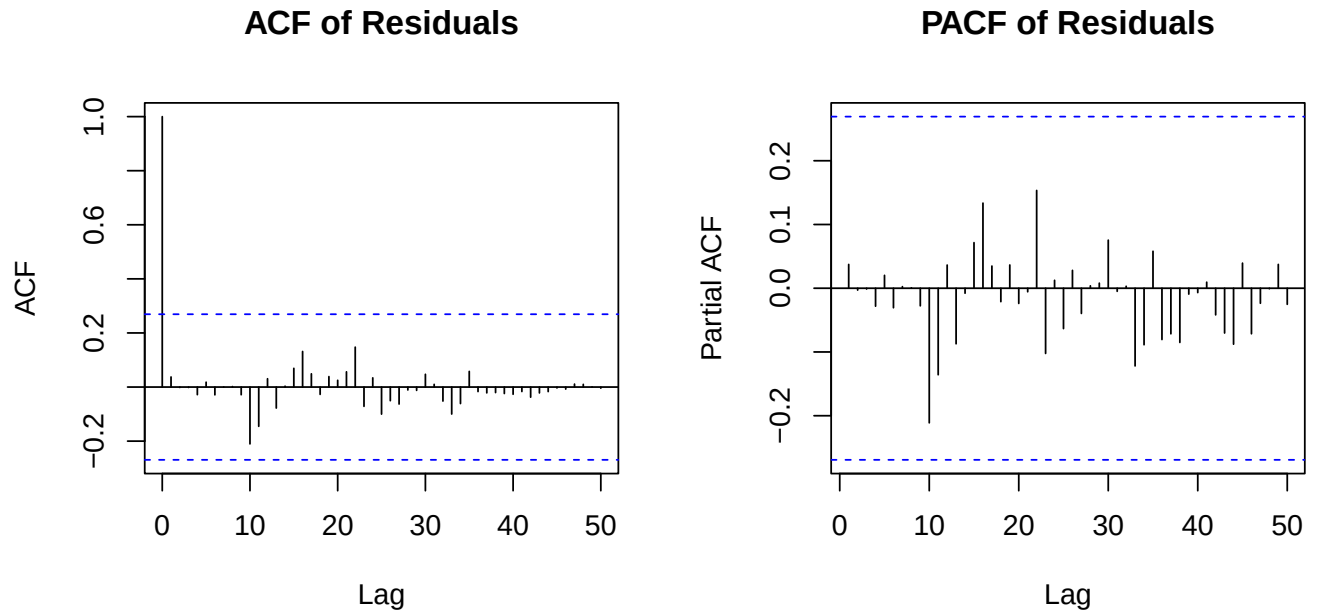
## Series: log(TLB_ts)
## ARIMA(15,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##    -0.1455 -0.1154  0.0919 -0.1862  0.0220  0.1559  0.1565  0.1742  0.0826
## s.e.   0.1315  0.1197  0.1028  0.0946  0.0935  0.0868  0.0972  0.0918  0.0931
##      ar10     ar11     ar12     ar13     ar14     ar15
##    -0.0706 -0.2325  0.2774 -0.3642 -0.3436 -0.4237
## s.e.   0.0939  0.0892  0.0988  0.1089  0.1264  0.1359
##
## sigma^2 = 0.00225: log likelihood = 88.78
## AIC=-145.56 AICc=-130.02 BIC=-114.34

```

Fitted ARIMA(15,1,0) model for TLB:

$$\begin{aligned}
& (1 + 0.1455B + 0.1154B^2 - 0.0919B^3 + 0.1862B^4 - 0.0220B^5 - 0.1559B^6 \\
& \quad - 0.1565B^7 - 0.1742B^8 - 0.0826B^9 + 0.0706B^{10} + 0.2325B^{11} - 0.2774B^{12} \\
& \quad + 0.3642B^{13} + 0.3436B^{14} + 0.4237B^{15})(1 - B) \log(y_t) \\
& = c + \varepsilon_t
\end{aligned}$$

Figure 29. ACF and PACF plots of residuals for ARIMA(15,1,0) model fitted to TLB



```

## Series: log(TLB_ts)
## ARIMA(12,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.3816 -0.3915 -0.4801 -0.1650 -0.0459  0.0622  0.0321  0.1705
## s.e.   0.1314   0.1535   0.1615   0.1495   0.1471   0.1454   0.1494   0.1589
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1587  0.0805  -0.0368  0.4897  0.5443  0.4755  0.9169
## s.e.   0.1683  0.1646   0.1479  0.1451  0.1011  0.2255  0.2718
##
## sigma^2 = 0.002424: log likelihood = 85.85
## AIC=-139.69  AICc=-124.15  BIC=-108.47

```

Fitted ARIMA(12,1,3) model for TLB:

$$\begin{aligned}
& (1 + 0.3816B + 0.3915B^2 + 0.4801B^3 + 0.1650B^4 + 0.0459B^5 - 0.0622B^6 \\
& \quad - 0.0321B^7 - 0.1705B^8 - 0.1587B^9 - 0.0805B^{10} + 0.0368B^{11} - 0.4897B^{12})(1 - B) \log(y_t) \\
& = (1 - 0.5443B - 0.4755B^2 - 0.9169B^3) \varepsilon_t + c
\end{aligned}$$

Figure 30. ACF and PACF plots of residuals for ARIMA(12,1,3) fitted to TLB

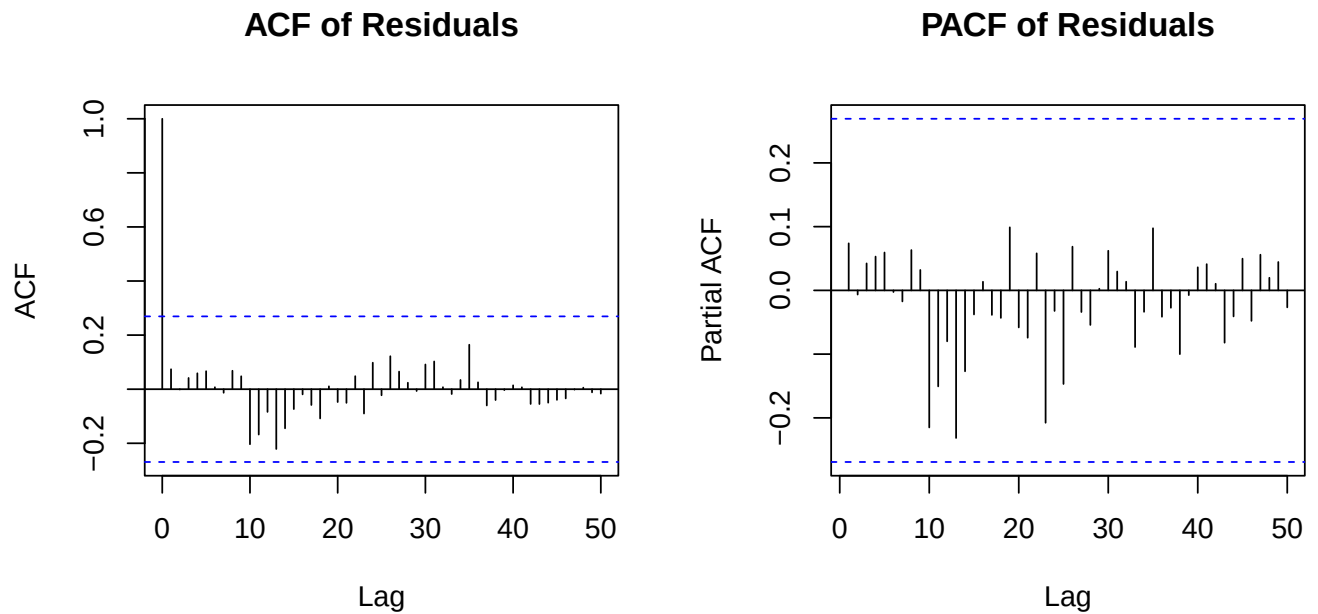


Table 5. Full table for TFR - no log transformation

Model	AIC	AICc	MSE	MAE	
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442	lowest AIC
ARIMA(16,1,1)	-62.39	-42.66	0.04783471	0.1700046	
ARIMA(15,1,1)	-60.26	-42.26	0.08431368	0.223524	
ARIMA(14,1,3)	-60.99	-40.26	0.01899351	0.09964934	
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547	lowest AICc
ARIMA(11,2,3)	-52.91	-39.19	0.05260825	0.1823268	
ARIMA(10,2,3)	-54.69	-43.02	0.05680021	0.1898403	

Table 6. Full table for TFR - log transformed

Model	AIC	AICc	MSE	MAE	
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592	lowest AIC Log Transform
ARIMA(12,1,4)	-131.83	-113.83	0.02101819	0.1132223	
ARIMA(13,1,5)	-133.96	-110.21	0.05826379	0.1840232	
ARIMA(12,1,5)	-130.02	-109.3	0.02172126	0.1147922	
ARIMA(15,2,0)	-133.67	-117.67	0.01869052	0.1056717	
ARIMA(15,2,2)	-135.54	-114.17	0.02235162	0.1187815	
ARIMA(14,2,2)	-10.49	-111.94	0.01087075	0.08352761	
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933	lowest MSE & MAE
ARIMA(12,2,2)	-127.34	-113.62	0.0498791	0.1720094	

Table 7. Full table for TLB - no log transformation

Model	AIC	AICc	MSE	MAE	
ARIMA(14,1,0)	975.33	988.66	67367429	6853.008	
ARIMA(14,1,1)	975.62	991.17	84179161	7769.997	
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072	lowest AICc (not log transformed)
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802	lowest AIC (not log transformed), MSE & MAE

Table 8. Full table for TLB - log transformed

Model	AIC	AICc	MSE	MAE	
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34	lowest AIC & AICc (log transformed)
ARIMA(15,1,1)	-145.43	-127.43	2173041601	46542.86	
ARIMA(14,1,1)	-140.56	-125.02	2211919788	46958.53	
ARIMA(13,1,1)	-141.45	-128.11	2200303964	46836.59	
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92	

Figure 1. Comparison of ARIMA Forecasts for TFR

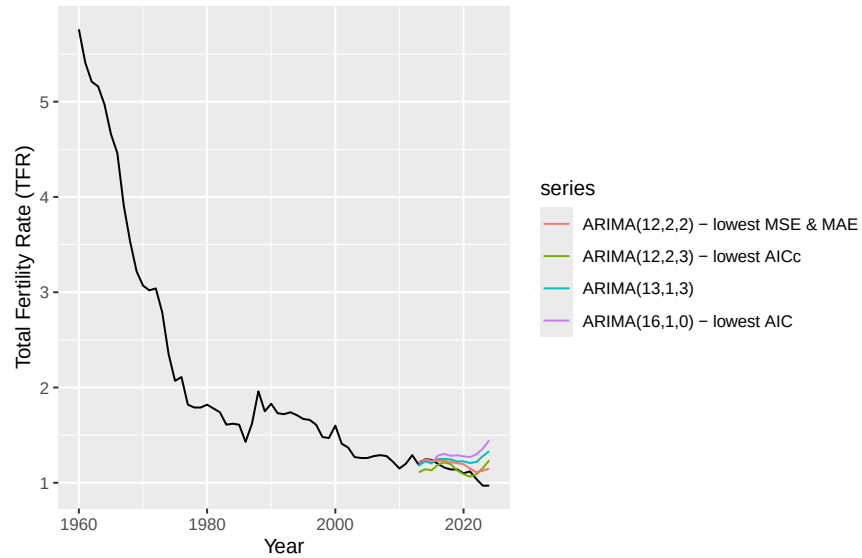


Figure 2. Comparison of ARIMA Forecasts for TLB

